

Zhiyu Gui

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Summary

Junior undergraduate student in Computer Science and Technology at the School of the Gifted Young, University of Science and Technology of China (USTC). GPA: 3.85/4.3 (Rank: 22/219). Specialized in LLM optimization, robotics, and autonomous systems development.

Education

University of Science and Technology of China (USTC) Sept 2022 – Jun 2026
Bachelor of Science in Computer Science and Technology(expected)
 ◦ GPA: 3.85/4.3, Rank: 26/215

Awards

Champion, RoboGame 2023 Robotics Competition Nov 2023
University of Science and Technology of China
 ◦ Responsible for robot vision, positioning, and communication.

Champion, Artificial Intelligence Innovation Application Competition Sep 2023
School of Information Science and Technology, USTC
 ◦ Solo project focusing on image semantic segmentation.

Silver Award, RWKV 2025 Ecosystem Content Collection Competition Jan 2025
Yuanshi Intelligence
 ◦ Developed the first RWKV backend inference framework supporting continuous-batching.

Projects

Count-R1: RL for LLM Counting Task Reasoning [Count-R1](#)
 ◦ Improved Qwen-3B accuracy (40%) via RL (no warm-start), observing emergent CoT.
 ◦ Verified RL needs strong base models; distillation outperformed RL on smaller models.

AttnInput: Context-Aware Pinyin Input Method [AttnInput](#)
 ◦ Enhanced Pinyin input method using large language models (LLMs).
 ◦ Responsible for backend model inference and training.

conRWKV: High-Concurrency RWKV Backend Inference Framework [conRWKV](#)
 ◦ First RWKV backend framework supporting continuous-batching and chunk prefill.
 ◦ Significantly reduced Time-To-First-Token (TTFT).
 ◦ Provided OpenAI-compatible API interface.

Skills

Backend Development: Python, C, C++, Cython, SQL

Frontend Development: PyQt

Libraries: PyTorch, Transformers, vLLM, FastAPI

GPU Kernel Development: Triton, CUDA, HIP

Hardware Development: Verilog, JLCEDA, Server Setup

Interests

LLM: LLM Reasoning, LLM Inference Optimization, Linear LLM Architecture Design, LLM Interpretability

Embodied Intelligence: Dexterous Hands, Visual-Tactile Sensors

Autonomous Driving: Inland Waterway Autonomous Navigation

Publications

AttnInput: Advancing Context-Aware Pinyin Input with Efficient Language Model Integration

Submitted to ACL2025

Zhiyu Gui, Xulei Sun

- Enhanced pinyin input method using large language models (LLMs), achieving state-of-the-art performance in abbreviated pinyin input while significantly reducing training costs.
- [Preprint available](#) 

Certificates

Kunpeng Ascend Training Camp Certificate

Jul 2024

Huawei

Work Experience

Intern

Jan 2025 – Feb 2025

Zhejiang Blue City Zhige Technology Co., Ltd.

- Participated in the development of autonomous navigation technology for inland new energy vessels.